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Week 9 Lab Assignment

Investigating the Predatory Stress Response of *Hemigrapsus oregonensis* to the *Carcinus maenas* in Warming Environments

Abstract

The Puget Sound ecosystem is being influenced by both invasive European Green Crabs (*Carcinus maenas*) and increasing temperatures associated with climate change. *Carcinus maenas* preys on native Hairy Shore Crabs (*Hemigrapsus oregonensis*) and eelgrass habitats, disrupting the natural state of the Puget Sound. The effects of *Carcinus maenas* on the Puget Sound ecosystem and other ecosystems similar to it have been intensely studied. However, there is a lack of research in studying how climate change could influence the prey and predator interactions of *Hemigrapsus oregonensis* and *Carcinus maenas*, especially since such organisms are known to be sensitive to changes in temperature. At the beginning of the study, it was predicted that there would be a significant difference between the statistics of *Hemigrapsus oregonensis* in ambient water and elevated water. Our study showed significantly increased physical activity from *Hemigrapsus oregonensis* in elevated water, with crab righting times and respiration remaining consistent among *Hemigrapsus oregonensis* in ambient and elevated water. The data itself suggests that temperature increases have little effect on *Hemigrapsus oregonensis*. However, the data is overall inconclusive, since flaws in the testing methods of *Hemigrapsus oregonensis* being in the presence of *Carcinus maenas* likely led to inconclusive data. Therefore, the data cannot be used to make a definitive conclusion on how the prey and predator interactions of *Hemigrapsus oregonensis* and *Carcinus maenas* are affected by climate change. Future research is recommended to study the physiological changes *Hemigrapsus oregonensis* and *Carcinus maenas* undergo, and how their prey and predator interactions change across varying temperatures.

Introduction

The Puget Sound ecosystem is known for its biodiversity and copious salmon harvests. Unfortunately, this biodiverse ecosystem is being degraded by foreign species and rapid changes in the environment. European Green Crabs (*Carcinus maenas*) are an invasive species of the Puget Sound area, posing a predatory threat to the native species of Hairy Shore Crabs (*Hemigrapsus oregonensis*) and eelgrass habitats. Ongoing research is investigating methods of removing *Carcinus maenas*, however there is little research investigating whether or not climate change is affecting the ecological disruption caused by *Carcinus maenas*, and its intensity (Fisher et al, 2024).

Climate change primarily affects the Puget Sound area by progressively increasing annual temperatures of the ecological systems (Walker et al, 2022). The effects of increasing temperatures on the Puget Sound ecosystem are vital, because temperature affects most ecological processes. In the case of studying *Carcinus maenas* and its ecological effects, temperature affects crab metabolic rate and potentially *Carcinus maenas* special distribution (Nielsen et al, 2025). Therefore, both *Carcinus maenas* and *Hemigrapsus oregonensis* are most likely affected by climate change, and by extension, their predatory and prey relationship. Studying the potential effects of increasing temperatures on the predator and prey relationship between *Carcinus maenas* and *Hemigrapsus oregonensis* is vital for understanding how ecosystems will change in accordance to climate change. Such information can be used to better aid conservation efforts of *Hemigrapsus oregonensis* and animals in similar conditions while understanding and adapting to the effects of climate change on the Puget Sound ecosystem and other similar ecosystems.

Methods

To study *Carcinus maenas* and *Hemigrapsus oregonensis* interactions across varying temperatures, the stress response of *Hemigrapsus oregonensis* when in the presence of *Carcinus maenas* was used to gauge the predator and prey interaction. Hemigrapsus *oregonensis* stress response was measured by respiration rate, distance travelled over time, and righting time after exposure to *Carcinus maenas*. This approach minimized *Hemigrapsus oregonensis* casualties to predation by *Carcinus* maenas, and allowed for measurement of the intensity of changes in metabolic rate in *Hemigrapsus oregonensis* when comparing two trials of varying temperatures. Crabs in elevated temperatures were expected to have different respiration rate, distance travelled over time, and righting times than crabs in ambient temperatures.

One rectangular tank filled with 14 C˚ water and another rectangular tank filled with 24 C˚ water were used to simulate conditions of ambient water temperature and elevated water temperature. The water at 14 C˚ is referred to as ambient water and the water at 24 C˚ is referred to as elevated water. Both temperatures of water were stored in bottles so they can be stored in containers which kept their respective temperatures. The waters would be poured back in the tanks once a set of trials began. A shrimp net was placed inside the tank near one of the thinner ends of each tank. Two rows of white blocks were placed parallel to each other roughly two inches apart and perpendicular to the shrimp net. The white block rows extended from the shrimp net to the other side of the tank.

A total of 12 *Hemigrapsus oregonensis* crabs were tested in this study, which were sampled from the shore of Bremerton, Washington state. They were divided into 2 groups of 6, with one group assigned to 14 C˚ ambient water and another assigned to with 24 C˚ elevated water. The groups were chosen to consist of 3 males and 3 females, and were stored in plastic containers, one was assigned to ambient water and the other was assigned to elevated water. Each container was filled and submerged in their respective water temperatures for the duration of the experiment.

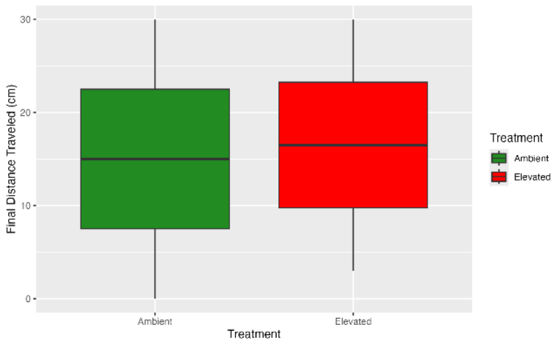
A trial consisted of one *Carcinus maenas* crab placed in the small area behind the shrimp net, opposite of the white block rows. Then, *Hemigrapsus oregonensis* crabs were placed on the side of the shrimp net between the while blocks. The *Hemigrapsus oregonensis* crabs were in the same water as the *Carcinus maenas* crab, which was designed to illicit a prey reaction out of the *Hemigrapsus oregonensis* crabs. Once placed inside a tank, a video for each crab was taken from the dorsal side of the tank to measure the crabs total distance traveled over time. The *Hemigrapsus oregonensis* crabs contained in ambient water were tested in the tank filled with ambient water and the *Hemigrapsus oregonensis* crabs contained in elevated water were tested in the tank filled with elevated water. After 30 seconds of filming, the trial concluded, and the *Hemigrapsus oregonensis* crab being tested was placed back in their respective container.

Then, each crab was taken to a flat countertop, where the crab was placed upside down and timed until it rightened itself to test their righting times. The crabs were placed back in their Then, 4 of the crabs of the ambient water and the elevated water were each placed in small sampling trays filled with resazurin. Every 15 minutes, the resazurin would be sampled over a 45-minute period. Once the samples are collected, the resazurin samples would be used to determine the level of metabolic activity of the tested *Hemigrapsus oregonensis* crabs. The tested *Hemigrapsus oregonensis* crabs would then be placed in a clear water tank for 5 minutes, and then placed in another clear water tank for 5 minutes before being placed in their original plastic container, to dilute the resazurin.

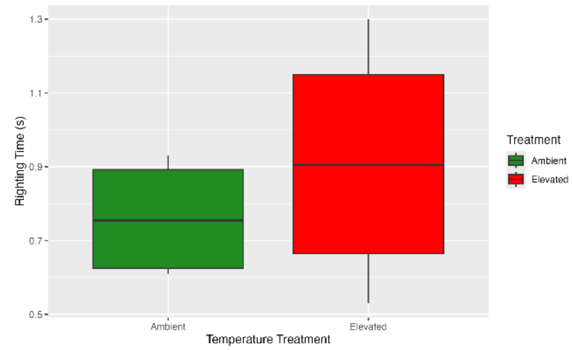
Each crab was tested 3 times total over a three-week period, which were called weeks six, seven, and eight. In the first week the *Hemigrapsus oregonensis* crabs were tested for distance traveled, righting time, and resazurin without any exposure to *Carcinus maenas*. In the second week, the *Hemigrapsus oregonensis* crabs were tested for distance traveled, righting time, and resazurin before exposure to *Carcinus maenas*, and then tested for distance traveled, righting time, and resazurin a second time after exposure to *Carcinus maenas*. The third final week tested the crabs once for distance traveled, righting time, and resazurin after exposure to *Carcinus maenas*.

Results

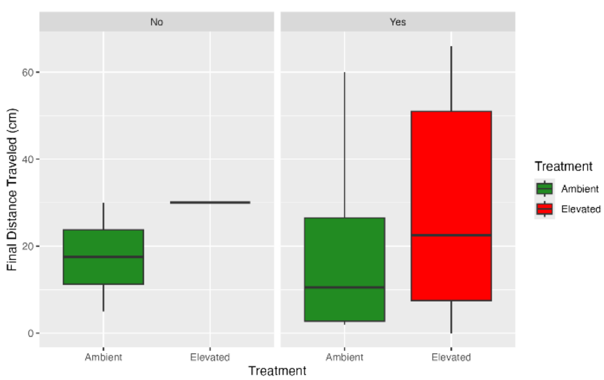
All crabs survived the sixth and seventh weeks. Three of the crabs in the elevated water died during week eight, with some of them dying during the testing trials. Therefore, the data for week 8 is incomplete. In week 6, the data presents the *Hemigrapsus oregonensis* crabs in ambient water and elevated water had equal final travel distance, and that crabs in elevated water had, on average, longer righting times than crabs in ambient water. In week 7, crabs in ambient temperature had an increased range of final travel distance after exposure to *Carcinus maenas*, and the crabs in elevated temperature had a drastically increased range of final travel distance, greatly exceeding the crabs in ambient temperature. In week 8, the remaining crabs in ambient temperature and elevated temperature had similar righting times. For all crabs which were tested, resazurin testing showed similar trends and respiration rates between *Hemigrapsus oregonensis* crabs in ambient water and elevated water. Respiration level across a 45-minute period rapidly increased during week 6 resazurin testing, and weeks 7 and 8 showed a trend of very little increase in respiration level.



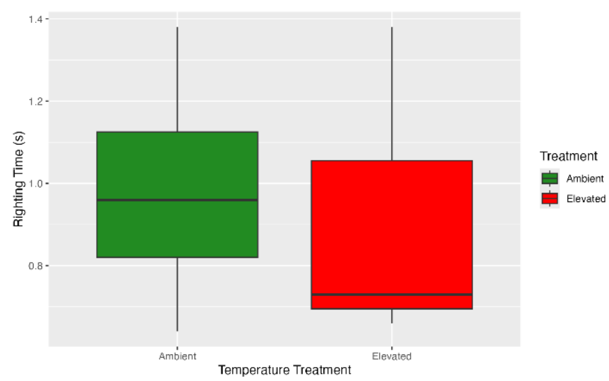
Ranges of final distance (cm) traveled by the *Hemigrapsus oregonensis* crabs during week 6 testing in ambient and elevated water.



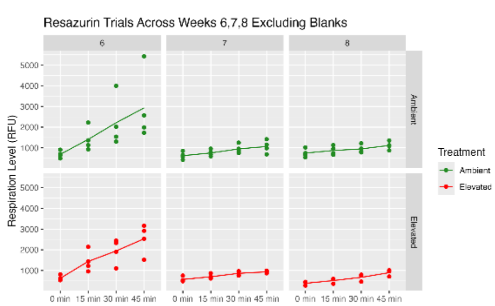
Ranges of righting time (sec) of *Hemigrapsus oregonensis* crabs during week 6 testing in ambient and elevated water.



Ranges of final distance (cm) traveled by the *Hemigrapsus oregonensis* crabs during week 7 testing in ambient and elevated water, tested with *Carcinus*



Ranges of righting time (sec) of *Hemigrapsus oregonensis* crabs during week 8 testing in ambient and elevated water prior to *Carcinus maenas* exposure



The respiration levels (RFU) of *Hemigrapsus oregonensis* crabs in ambient and elevated water temperatures over weeks 6, 7, and 8, calculated by resazurin samples taken in 15-minute intervals over a 45-minute period.

Discussion

The data collected shows a trend of *Hemigrapsus oregonensis* crabs in elevated water temperatures travelled more, and yet had similar righting times as *Hemigrapsus oregonensis* crabs in ambient water. Crabs in both ambient and elevated water had similar respiration levels across weeks 6, 7, and 8. Other than increased rates of travel exhibited from crabs in elevated water, there is very little difference between the two tested populations of *Hemigrapsus oregonensis.* According to the data alone, elevated temperatures appear to have very little effects on *Hemigrapsus oregonensis* crabs, therefore, making a primary effect of climate change insignificant for their conservation. However, it is worth noting that 3 of the 6 *Hemigrapsus oregonensis* crabs in elevated temperatures died as all 6 *Hemigrapsus oregonensis* crabs in ambient temperatures survived. Therefore, increased temperatures may have some negative effect on *Hemigrapsus oregonensis,* however there was too little data and too small of a sample to make a significant conclusion. Therefore, the data collected so far is inconclusive. There were also many flaws with how the *Hemigrapsus oregonensis* crabs were tested. During the flipping testing, the *Hemigrapsus oregonensis* crabs would either flip forwards over their front claws or backwards. Crabs which happened to flip backwards had significantly shorter righting times than crabs which flipped forwards, so the data was likely influenced by the inconsistent methods of how the crabs flipped themselves. Also, crabs in the same container had different amounts of usable legs, and died during testing, so the data was likely skewed by such physiological differences. The change in increasing rates of respiration in the resazurin treatment Can be the result of an error on week 6 or the tested *Hemigrapsus oregonensis* crabs becoming conditioned to the resazurin.

Previous research focused on the effects of *Carcinus maenas* on the Puget Sound ecosystem, and how temperature affects crab distribution. This study couldn't provide supported conclusions for how climate change factors in to the disruption of *Carcinus maenas*, so future studies should focus on how all species of Puget Sound are directly affected by climate change, and how their behavior and species interactions change, so conservation efforts could adapt to climate change's effects.

References

Fisher, M. C., Grason, E. W., Stote, A., Kelly, R. P., & Litle, K., McDonald, P. S. (2024). Invasive European green crab (Carcinus maenas) predation in a Washington State estuary revealed with DNA metabarcoding. *PLoS ONE*, 19(5): e0302518. doi:10.1371/journal.pone.0302518.

Nielsen, M. S., Svendsen, J. C., Wilms, T., Bertelsen, J. L., Kruse, B. M., & Lindegren, M. (2025). Factors influencing the abundance of European green crab (Carcinus maenas): Combined effects of temperature, habitat and predator release. Estuarine, *Coastal and Shelf Science*, 322, 109374. doi:10.1016/j.ecss.2025.109374.

Walker, S., Mozaria-Luna, H. N., Kaplan, I., & Petatán-Ramírez, D. (2022). Future temperature and salinity in Puget Sound, Washington State, under CMIP6 climate change scenarios. *Journal of Water and Climate Change*. 13 (12): 4255–4272. doi: 10.2166/wcc.2022.282.